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USER MANUAL

September 29, 2025

North Shore Cougar Habitat Suitability Calculator

```
==========================================================================
  |     |   North Mountain Cougar Habitat Suitability Analysis
  |     |     |    6 of S  Polygons in Study Area
==========================================================================
Low suitability:
  |     |     |       Number of polygons:     2
  |     |     |     Minimum polygon area:   340.000
  |     |     |     Maximum polygon area:   400.000
  |     |     |               Total area:   740.000
  |     |     |     Average polygon area:   370.000

Medium suitability:
  |     |     |       Number of polygons:     3
  |     |     |     Minimum polygon area:   148.000
  |     |     |     Maximum polygon area:   235.000
  |     |     |               Total area:   533.000
  |     |     |     Average polygon area:   177.667

High suitability:
  |     |     |       Number of polygons:     1
  |     |     |     Minimum polygon area:   105.000
  |     |     |     Maximum polygon area:   105.000
  |     |     |               Total area:   105.000
  |     |     |     Average polygon area:   105.000
==========================================================================
```

Overview

The North Shore Cougar Habitat Suitability Calculator is an app that takes data on a collection of 4 to 10 forest stands, categorizes them as being low, medium, or high suitability areas for being habitats for the north shore cougar, then outputs general information about the amount of area covered by the inputted forest stands.

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Parameters

The North Shore Cougar Habitat Suitability Calculator requires 6 parameters:

- Dominant tree species
- Number of forest stands
- Coverage area of forest stands
- Tree diameter in centimetres
- Tree height in metres
- Cover type

These parameters are either used to calculate habitat suitability, or included as part of the final report. Details on each parameter can be found below:

Dominant Tree Species

The dominant tree species parameter accepts a code indicating the dominant tree species for all inputted forest stands. The prompt for inputting the dominant tree species appears as such:

```
Please enter one of the following dominant tree species code:
  AP - Austrian Pine
      JP - Jack Pine
       RP - Red Pine
     SP - Scots Pine
     WP - White Pine
   BS - Black Spruce
  NS - Norway Spruce
     RS - Red Spruce
   SS - Sitka Spruce
   WS - White Spruce
     Enter the code:
```

Of tree types, only those that are either a pine tree or a spruce tree are considered. Inputting the tree's full name is not accepted: only the code. The response given is **not used to calculate** habitat suitability and is instead displayed in the final report. Information pertaining to species codes can be found in the 'Nova Scotia Interpreted Forest Inventory - Current Forest Data'.

Number of Forest Stands

The number of forest stands parameter accepts a number between 4 and 10 and indicates the number of distinct forest stands that should be evaluated. The prompt for inputting the number of forest stands appears as such:

```
How many forest stands do you want to examine?:
```

The chosen value will change how many stands the user is asked for data for, and how many listed in the final result. Please be sure of how many stands need to be examined before inputting a number. Entering a number less than 4 will default to 4, and a number greater than 10 will default to 10.

Coverage Area of Forest Stands

This, and each of the following parameters, will require the user to input a value for each individual stand, as opposed to broadly for the whole dataset. Coverage area of forest stands accepts any whole number as input, but is designed for a 3 digit number or less. Input should describe the area covered by the indicated forest stand. Entering a unit of measurement is not supported. The prompt for inputting the coverage area of a forest stand appears as such:

```
Enter value for the area of stand  1:
```

The given value is **not used to calculate** habitat suitability, but does alter data displayed in the final report.

Tree Diameter in Centimetres

The tree diameter in centimetres parameter accepts any whole number as an input. It expects a value for the average diameter of trees in a given forest stand and uses it as part of the calculation for habitat suitability for the north shore cougar. The prompt for inputting the tree diameter in centimetres is as follows:

```
Enter value for the average tree diameter in centimetres of stand  1:
```

Input for this value must be repeated for each individual forest stand.

Tree Height in Metres

The tree height in metres parameter accepts any whole number as an input. It expects a value for the average height of trees in a given forest stand and uses it as part of the calculation for habitat suitability for the north shore cougar. The prompt for inputting the tree height in metres is as follows:

```
Enter value for the average tree height in metres of stand  1:
```

Input for this value must be repeated for each individual forest stand.

Cover Type

The cover type parameter allows one of three distinct inputs to be entered: softwood, mixedwood, or hardwood. This parameter is used to identify what the primary form of cover is within a forest stand. The prompt for input appears as such:

```
Enter one of the following cover types for stand 1:
     Softwood - S
     Mixedwood - M
     Hardwood - H
 Enter the code:
```

The app will take the first letter of whatever text is inputted and accept it as the input, so be sure the first letter is one of ‘S’, ‘M’ or ‘H’. It is **not** case sensitive. The given input is used as part of the calculation for habitat suitability and must be repeated for each individual forest stand.

Calculating Suitability

Suitability of each forest stand as a habitat for the north shore cougar is calculated using three metrics: average tree height, average tree diameter, and coverage type. Depending on the stated value of a parameter for a forest stand, the suitability rating value of the stand is increased by a distinct amount, as shown below:

AVERAGE TREE DIAMETER (AVDI)	
Category	**Rating Value**
Less than 20 centimeters	0.75 points
Between 20 and 30 centimeters	1.75 points
Greater than 30 centimeters	2.5 points
FOREST STAND AVERAGE TREE HEIGHT (HEIGHT)	
Category	**Rating Value**
Less than 10 meters	1.25 points
Between 10 meters and 20 meters	2.5 points
Greater 20 meters	3.75 points
COVER TYPE (COVER_TYPE)	
Category	**Rating Value**
Softwood	1 point
Mixedwood	2 points
Hardwood	3.75 points

These values are added up to produce an overall suitability rating value. For example, if a forest stand had trees with an average diameter less than 20 centimetres, between 10 and 20 metres, and the cover type was mixedwood, the suitability rating value would be 0.75 + 2.5 + 2, to equal 5.25. Once a suitability rating value is assigned to a stand, it is then categorized as being one of either low, medium, or high suitability as a habitat for the north shore cougar. That categorization process is based on the following values:

Suitability Rating	
Category	**Rating Value**
Low Suitability	< 5
Medium Suitability	Between 5 and 8
High Suitability	>8

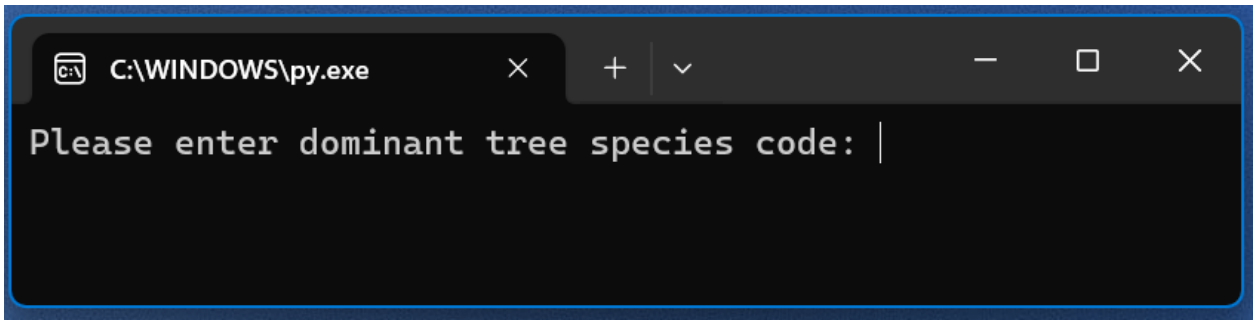
If we were to examine the previous example, we would see that its value of 5.25 lands between 5 and 8, thus giving the forest stand an overall rating of medium suitability. This process is repeated for each individual forest stand in the dataset.

Operating Instructions

Here are step-by-step instructions for using the North Shore Cougar Habitat Suitability app.

Step 1: Opening the App

To open the North Shore Cougar Habitat Suitability app, you'll need to have version 3.10 downloaded from https://www.python.org/downloads/ Earlier or later versions may not work as intended. Once installed, find where the app is stored on your device, right click it, select open with, then choose where python 3.10 is stored within your device. You should now have a pop-up window that resembles what is shown here:



Step 2: Inputting Data

Here you will be prompted for data related to the study area. The following section will give a brief overview of accepted values: more information about each parameter can be found in the 'Parameters' section of this user manual.

Step 2.A

Start by entering the code for the dominant tree species within the study area, as per the 'Nova Scotia Interpreted Forest Inventory - Current Forest Data'. For this example we will use WI for willow.

```
Please enter dominant tree species code: WI
```

Hit enter.

Step 2.B

Next, input the number of forest stands that will be included in the habitat suitability calculation, between 4 and 10. For this example we will look at 4 forest stands.

```
How many forest stands do you want to examine?: 4
```

Hit enter.

Step 2.C

You will now be prompted to enter the average tree diameter in centimetres for each of your forest stands. Henceforth, it is important to **keep note of the forest stand** you are currently inputting data for, as stated at the end of the prompt. Inputting data from the wrong stand during any one of these steps will taint the final results.

```
Enter value for the average tree diameter in centimetres of stand  1:23
Enter value for the average tree diameter in centimetres of stand  2:18
Enter value for the average tree diameter in centimetres of stand  3:34
Enter value for the average tree diameter in centimetres of stand  4:11
The numbers entered are :  [23, 18, 34, 11]
```

Be sure to hit enter after each entry. Repeat the steps for both ‘average tree height’ and ‘area of stand’.

Step 2.D

Finally input the cover type code for each stand. Acceptable values are ‘S’ for softwood, ‘M’ for mixedwood, or ‘H’ for hardwood. Like Step 2.C, be sure you are inputting values for the correct stand.

```
Enter one of the following cover types for stand 1:
     Softwood - S
     Mixedwood - M
     Hardwood - H
 Enter the code: S
```

Hit enter after each entry.

Step 3: Understanding the Results

After you have inputted all required data, you'll see a report that appears similar to the one imaged below:

```
==========================================================================
  |     |   North Mountain Cougar Habitat Suitability Analysis
  |     |     |    4 of WI Polygons in Study Area
==========================================================================
Low suitability:
  |     |     |       Number of polygons:     2
  |     |     |     Minimum polygon area:    31.000
  |     |     |     Maximum polygon area:   134.000
  |     |     |               Total area:   165.000
  |     |     |     Average polygon area:    82.500

Medium suitability:
  |     |     |       Number of polygons:     1
  |     |     |     Minimum polygon area:   543.000
  |     |     |     Maximum polygon area:   543.000
  |     |     |               Total area:   543.000
  |     |     |     Average polygon area:   543.000

High suitability:
  |     |     |       Number of polygons:     1
  |     |     |     Minimum polygon area:   351.000
  |     |     |     Maximum polygon area:   351.000
  |     |     |               Total area:   351.000
  |     |     |     Average polygon area:   351.000
==========================================================================
```

The app has categorized each stand as being one of low, medium or high suitability as a habitat for the north shore cougar. Within each of those classifications, it specifies how many stands were designated the corresponding rating, the smallest and largest stand with that classification, the sum total area of all stands and the average area each stand covers. More information on how these classifications are reached can be found in the 'Calculating Suitability' section of this user manual.

Step 4: Repeating the Process

After receiving the report, you'll be asked if you'd like to classify another set of stands. Responding with the letter 'N' will close the app, all other responses will run the analysis again. Only the first letter in the entry is considered.

Usual Errors

The following is a list of all built-in error codes and what they mean. The * symbol will be used in lieu of a value that can vary depending on user input.

- Incorrect input. Please enter one of the following dominant tree species code:

Tree species code requires an input that starts with one of the listed tree species codes. Please note that only codes for pine or spruce trees are allowed, please enter the species code and not the full name.

- * Is too small, the number will be 4
- * Is too many, the number will be 10

The app only operates with 4 to 10 forest stands at a time. Entering a number smaller or larger than that will cause the app to default to the smallest or largest accepted value, respectively.

- Enter a number please.

A value has been entered that is not a whole number. A whole number must be entered.

- Invalid value for the average tree diameter in centimetres of stand *
- Invalid value for the average tree height in metres of stand *
- Invalid value for the area of stand *

The input for these parameters must be a whole number: entries with decimals or non-number characters will not be accepted.

- Invalid input. Please enter one of the following cover types for stand *

Cover type is a parameter that only allows for one of three cover types: softwood, mixedwood, or hardwood. Choose one of these options by typing one of the letter ‘S’, ‘M’ or ‘H’, respectively.

Limitations

The North Shore Cougar Habitat Suitability Calculator app does well to rate the suitability of forest stands as north shore cougar habitat and provide a general overview of its analysis. That said, there are limitations to its capabilities, some of which are written below:

- The app does not accept less than 4 or more than 10 stands at one time.
- It cannot accept data files such as .csv or .xsl
- It only accepts whole numbers as values for area, tree height, or tree diameter.
- The dominant tree species must be the same for all forest stands.
- The dominant tree species can only be a type of pine or spruce.
- It does not take into consideration all variables that may impact a stand's suitability as habitat land for the north shore cougar.
- The final report is limited in the data that it displays.

This list does not claim to be a complete set of all limitations of the app and other limitations may exist that are not included here.